Algorithm 1 Statics of one continuum leg

```
Initialize q^0
for i = 1, ..., maxiter do
    Form the residual:                          ▷ simulate states and gradients
```

$$\boldsymbol{b} \leftarrow \boldsymbol{F}(\boldsymbol{q}^{i-1}) + \boldsymbol{M}\boldsymbol{g} \tag{3}$$

```
    Compute A(q^{i-1}) = ∂F(q^{i-1}) / ∂q
    Find dq:                                    ▷ Newton solve
```

$$\boldsymbol{A}\, d\boldsymbol{q} = \boldsymbol{b} \tag{5}$$

```
    q^i ← q^{i-1} + α dq                        ▷ take solver step
    if ‖q^i − q^{i-1}‖ < ε then
        return q^i                              ▷ solver converged
    end if
end for
```